Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **developer-facing APIs and automation**, **C++/Python systems**, cloud services, and reliable software spanning distributed components, simulations, and real-world operational workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant FastAPI services with explicit job states, timeouts, failure isolation, and monitoring, processing 1K+ jobs at 3x throughput while strengthening **reliable automation across distributed production workflows**.
- Architected multi-tenant Azure SQL and cloud-storage integrations for 500+ users, reducing dashboard latency 71% through **indexed queries, validation, and maintainable API boundaries** across customer-facing platform services in production.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask workflows for secure uploads, processing history, model controls, and feedback, supporting 10K+ daily requests through **REST APIs, production debugging, and user-focused iterative delivery**.
- Designed concurrent Python automation with ThreadPoolExecutor, OCR, timeouts, and model fallbacks, processing 1K+ files per batch while reducing turnaround 60% through **testing, validation, and fault isolation during peak loads**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a PyTorch Decision Transformer for autonomous-driving simulation, processing 100K+ trajectory logs through reproducible pipelines and improving planning 18% across **controlled software-to-physical-system research workflows for autonomy experiments**.
- Integrated HD-map geometry, lane direction, and ego trajectories using spatial-temporal encoding and multi-head attention, improving closed-loop stability 15% while **debugging complex interactions across simulation and control components**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Engineered cross-platform APIs and object-oriented modules using Singleton and Factory patterns, improving response times 30% while **debugging integrations reliably across Windows, Linux, macOS, AWS Glue, and S3**.
- Automated builds and 20+ unit and integration suites with Jenkins, PyTest, Mockito, and virtual machines, increasing pipeline efficiency 68% through **CI/CD, reproducible validation, and multi-platform release automation**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, React interfaces, and reliable real-time workflows** across distributed locations.
- Automated Docker packaging and Jenkins delivery, cutting deployment time 50% while using Sentry, Azure Monitor, and Kibana for **observability, structured debugging, and dependable production support** across services.

Skills

Languages: C++, Python, TypeScript, JavaScript, Java, SQL, Bash, HTML5, CSS3
APIs & SDKs: REST, gRPC, OpenAPI, AsyncAPI, SDK Design, API Versioning, Protocol Buffers, Webhooks, FastAPI, Flask
Automation & Devices: MQTT, NATS, Device Control, Embedded Systems, Networking, Telemetry, State Machines, Industrial Automation, MES
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Git, Linux
Engineering: OOP, Data Structures, Algorithms, Distributed Systems, Unit Testing, Integration Testing, Debugging, Monitoring, Logging, Reliability, Fault Isolation, Documentation, AI Applications

Projects

PrintFlow SDK | *C++, Python, REST, gRPC, OpenAPI, Protocol Buffers*

- Designed a **developer-facing printer SDK** with typed C++ and Python clients, REST/gRPC interfaces, OpenAPI contracts, idempotent job submission, versioned resources, actionable errors, and reliable asynchronous print-status callbacks.

Device Automation Lab | *C++, Python, MQTT, NATS, Docker, PyTest*

- Built a **device-automation simulator** using MQTT and NATS messaging, command state machines, telemetry, offline queues, reconnect policies, fault injection, and deterministic tests for reliable software-controlled manufacturing workflows.

CPU-Scheduling Simulator | *C++, React, JavaScript, Algorithms* | Live

- Implemented **FCFS, SJF, SRT, Round Robin, priority, and multilevel scheduling** in C++, validating correctness and visualizing execution behavior across 1K+ simulated process combinations with interactive React controls.